Prashanth Kumar Kadasi

AI Security Engineer — Agentic Systems, MCP Tool Permissions, LLM Guardrails

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SUMMARY

Engineer who builds and audits agentic/LLM systems end to end: threat modelling of tool-calling and MCP trust boundaries, source-code audit and responsible disclosure, and Python guardrails — permission boundaries, scope limits, budget enforcement, hash-chained audit ledgers — verified by independent evaluation harnesses.

FEATURED PROJECTS

BUGAGENTS — AUTONOMOUS VULNERABILITY RESEARCH

- 3 submission-ready vulnerability reports (two High), produced over 53 autonomous turns and 3,156 tool calls.
- 34 filter refusals before promotion; 8/8 harness tests pass; $0.067 spent against a $50 cap (budget enforcement).
- Source-code audit, proof-of-concept development, CVSS scoring, CWE classification, scope handling, responsible disclosure.

AGENTSWARM — MULTI-AGENT GUARDRAILS & AUDIT LEDGERS

- 32 verified runs, 32/32 hash-chained audit ledgers intact across 2,592 tool calls.
- 0 oracle violations across 1,254 tool calls; 0 of 2 self-assessing agents overclaimed results.
- 195/224 agent-written tests passing after independent re-run; 209 artifacts (4.1 MB) produced.
- Scope guardrails, bounded autonomy, human-in-the-loop escalation, budget enforcement.

AI ECO — MCP TOOLING & ENTERPRISE AI REVIEW

- 10 swarm agents on 15+ MCP tools and 5 resources; 83.4% fitness score.
- Living memory compaction bounded to <4,000 words; RAG, FastMCP, tool/function calling.

AWAKENING — LONG-HORIZON AUTONOMY & CI INTEGRITY

- 1,441 autonomous turns and 1,553 commits; sustained a 330-turn API blackout at 100% CI integrity.

JOBAGENTS — RAG PIPELINE & ANTI-FABRICATION REVIEW

- Sourced 7,533 jobs across 5,784 distinct roles; computed 22,647 match scores with 0 model calls.
- Anti-fabrication guard catches 8/8 seeded fabrications; 11 review invariants, 0 failures; score threshold 65; 1,864 duplicate postings merged.

MODEL EVALUATION & STEERABILITY

- BrandXY: 76.5% vs 25.5% baseline (+51% lift) with GPT-OSS-20B on AMD MI300X.
- pRash: benchmarked 6 autonomous coding CLIs over 2 rounds; multi-model A/B evaluation, bias quantification.

EXPERIENCE

Black Piano — Data Analyst / AI Systems | Remote | Mar 2026 – Present

- Build analytics automation and AI systems tooling in Python, SQL, and GCP (Cloud Run, Docker, GitHub Actions).

Pi Software Solutions (Pi - Datametrics) — Data Analyst | Remote | Mar 2023 – Feb 2026

- Delivered 18 ETL pipelines, 15+ enterprise dashboards, and 30+ client reports across BigQuery, Looker Studio, and Tableau.

EDUCATION

M. Pharmacy (Pharmaceutical Analysis & QA), JNTUH, May 2012.

SKILLS

- **Agentic:** Multi-Agent Swarms, FastMCP (MCP), Tool/Function Calling, RAG, Bounded Autonomy, Scope Guardrails, Budget Enforcement, Hash-Chained Audit Ledgers, Human-in-the-Loop Escalation, Long-Horizon Agent CI/CD, Agent Evaluation Harnesses, Multi-CLI Orchestration
- **LLM:** LoRA/QLoRA Fine-Tuning, Model Evaluation & Steerability, Multi-Model A/B Evaluation, Independent Verifier Design, Prompt Engineering, Bias Quantification
- **Security:** Source-Code Audit, Vulnerability Research, Proof-of-Concept Development, CVSS Scoring, CWE Classification, Responsible Disclosure
- **Languages:** Python (PyTorch, Pandas, NumPy, FastAPI), TypeScript, Node.js, SQL, JavaScript, HTML
- **Data & Infra:** BigQuery, Looker Studio, Tableau, ETL Orchestration, Data Modeling, Data Quality & Validation, GCP, Cloud Run, Docker, GitHub Actions, Linux/Bash, Git
- **Platforms:** Claude 4.x, Gemini 2.5/3.x, GPT-OSS 120B/20B, DeepSeek 4.1, Ollama, NVIDIA NIM, OpenRouter